

1.1 Dataset Requirements

- **At least 1,000 observations.**
 - The dataset has 30,000 observations of credit card clients in Taiwan, collected between April and September 2005.
- **At least 10 input variables/features before cleaning and reduction. Make sure that the dataset has a mix of both numeric and categorical variables.**
 - There are 23 input variables/features before cleaning and reduction. The dataset includes numerical, categorical, and ordinal features:
 - **Numerical features:** LIMIT_BAL, age, six monthly bill amounts, and six monthly payment amounts.
 - **Categorical features:** sex, marital status (MARRIAGE as shown in dataset), and education.
 - **Ordinal features:** six monthly repayment-status codes (PAY_0, PAY_2 to PAY_6 as shown in dataset).
- **A clearly identifiable target variable for a supervised classification or regression problem.**
 - Target variable: default payment next month (1 = default, 0 = no default). The classes are imbalanced, as about 22% of clients default.
- **A publicly accessible source, with the dataset URL and licence or terms of use identified.**
 - **Dataset:** Default of Credit Card Clients
 - **Source:** UCI Machine Learning Repository
 - **Creators:** I-Cheng Yeh and Che-hui Lien (2009)
 - **URL:** <https://archive.ics.uci.edu/dataset/350/default+of+credit+card+clients>
 - **Licence:** [Creative Commons Attribution 4.0 International](#)
- **The dataset should not already be so heavily cleaned or reduced that little meaningful preparation remains to be done.**
 - The dataset is not pre-cleaned. Preparation will focus on handling invalid/undocumented values and duplicates, skewness and outliers, and data redundancy. There are no missing values.
 - There is no direct target leakage, as all the features come before the target month. Preprocessing will still be fitted on the training split only.
- **Dataset off Kaggle not allowed.**
 - The dataset comes from the UCI Machine Learning Repository, not Kaggle.

1.2 Problem Definition

- **Define the target (dependent) variable and the predictor (independent) variables.**
 - **Target:** default payment next month (1 = default, 0 = no default).
 - **Predictors:** the 23 features above. Drop the ID column because it is an identifier.
- **State whether the intended task is classification or regression and justify the choice.**
 - Binary classification. The default is a yes/no event, and the dataset has no continuous outcome that a regression problem could target.
- **Explain the real-world relevance of the prediction problem.**
 - Financial institutions must decide how much credit to lend to a customer, but their main concern is the uncertainty about whether the customer will pay back on time. When defaults occur, lenders lose money; however, refusing good customers also loses them revenue.
 - Understanding which customer characteristics and repayment behaviours are linked to default helps institutions assess risk, identify customers more likely to default, and make better-informed credit decisions.
 - Our project looks at a customer's credit limit, age, personal details, and previous payment history to see whether this information can identify customers who may miss their payment in the upcoming month. Our focus is not to build the final model, but to understand and prepare the data and reduce the feature space before modelling.